

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: II

Problem Set 3

(Ancillary Statistics, Basu's Theorem, CRLB)

1. Let $X_1, \dots, X_n \sim U(\theta, \theta + 1)$. Show that the range $R = X_{(n)} - X_{(1)}$ is an ancillary statistic.
2. Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$. Find the CRLB for an unbiased estimator of p .
3. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$, where σ^2 is known. Define the sample mean $\bar{X} = \frac{1}{n} \sum X_i$ and the sample sum of squares $S^2 = \sum (X_i - \bar{X})^2$.
 - (a) Show that S^2 is ancillary for μ .
 - (b) Show that $\bar{X}$ is complete sufficient for μ .
 - (c) Prove that $\bar{X}$ and S^2 are independent.
4. Let $X_1, \dots, X_n$ be a random sample from $N(\theta, 1)$. Define the sample mean $\bar{X} = \frac{1}{n} \sum X_i$.
 - (a) Find the Cramér-Rao Lower Bound (CRLB) for estimating $g(\theta) = \theta^2$.
 - (b) Show that the UMVUE of θ^2 does not attain the CRLB.
 - (c) Find the CRLB for θ and determine if $\bar{X}$ attains this bound.
5. Let $X_1, \dots, X_n$ be a random sample from $\text{Exp}(\lambda)$. Define $T = \sum X_i$ and $U = \frac{X_1}{T}$.
 - (a) Show that T is complete sufficient and U is ancillary. Conclude their independence.
 - (b) Find the CRLB for an unbiased estimator of λ .
 - (c) Define $\theta = 1/\lambda$. Find the Fisher information in θ and find the CRLB for estimating θ .